

Aneesh Deshmukh

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in aneesh-deshmukh

I am currently pursuing a Master's degree in Computer Science, focusing on AI and robotic solutions for task and path planning inspired by neuroscience. I also serve as a Research Assistant at the P&G Digital Accelerator @ UC. My strengths include problem-solving, programming, and a strong foundation in deep learning and robotics. I am driven to apply these skills to real-world challenges through research and teaching.

EDUCATION

Masters of Science in Computer Science	GPA: 4.0
University of Cincinnati (UC)	Aug 2024 - May 2027 (<i>expected</i>)
Bachelor of Technology in Electronics and Communications Engineering	CGPA: 8.72
Vellore Institute of Technology (VIT) - Vellore	July 2017 - June 2021

SKILLS

Deep Learning, LLMs, Neural Networks, PyTorch, Keras, OpenCV, MATLAB, ROS, Isaac Sim, DSA, Javascript, Java, Python, C/C++, Google Cloud, Kubernetes, AWS, MongoDB, SQL, Redis, PostgreSQL, Database and Server Maintenance

PROFESSIONAL EXPERIENCE

P&G Digital Accelerator @ UC
Graduate Research Assistant
Stack: Python
September 2025 - Present

- Developed a Python/Dash dashboard to analyze experimental data and support decision-making workflows
- Built automation tools to streamline experimental data processing and reporting
- Designing active learning-based approaches to improve experimental design and sampling efficiency
- Developing robotic solutions to automate experimental workflows and enable rapid iteration

Apricity Robotics
Software Engineer Intern
Stack: Nvidia Isaac Sim, Isaac Labs, ROS, PyQt5, Python
June 2025 - August 2025

- Developed simulation pipelines in Isaac Sim and ROS2 for robotic sonography solutions.
- Integrated Python-based controllers for semi-autonomous robotic arm manipulation in diagnostic procedures.
- Contributed to system design for AI-enhanced echocardiography using robotic-assisted motion planning and sensor fusion.
- Collaborated on healthcare-focused robotic automation aimed at improving ultrasound imaging precision and workflow.

Metasports Media Pvt Ltd (previously - Octathorpe Web Consultants)
Stack: Javascript, Python, MongoDB, Redis, Docker, Google Cloud, Kubernetes, AWS, C#
Software Engineer
June 2021 - July 2024

- Developed a modular automation system reducing development time by 70% for core revenue products. Led cross-functional teams of 4–7 to scale systems and boost code quality from 10% to 64%.

- **Software Developer Intern**
Dec 2020 - June 2021
Built a ReactJS-based internal analytics dashboard for 100% product visibility. Handled backend feature deployment and production server maintenance (NodeJS, PHP, Ruby).

PUBLICATIONS

- **GuessingGame: Measuring the Informativeness of Open-Ended Questions in Large Language Models.** *EMNLP 2025*
Dylan Hudson, Daniel Vennemeyer, **Aneesh Deshmukh**, Justin Zhan, Tianyu Jiang.
Independently developed a Bayesian belief-tracking framework to measure per-question information gain in open-domain dialogue. This formed one of the core technical contributions of the

submission, enabling interpretable evaluation and improved LLM prompting strategies via soft evidence updates over semantic concepts.

RESEARCH

- **Uncertainty of thought (NLP/LLM)** with Dr Tianyu Jiang (University of Cincinnati). Having established multi-round context and belief tracking. We are currently focusing on analyzing an LLM's understanding of various costs of information retrieval approaches. I am also working on using Bayesian belief-tracking to further improve information retrieval.
- **Hippocampal Navigation using Neural Networks** in 3D space with Dr Ali Minai (University of Cincinnati). My research explores **embodied agents in visually diverse environments** that must learn spatial layouts and identify canonical entities using **vision-language models (VLMs)**. Specifically, I study how agents can explore environments like kitchens, interpret high-level natural language instructions, and execute goal-directed behaviors by inferring admissible entity sequences. My broader interest includes enabling generalization across environment classes and developing hierarchical spatial reasoning across multiple rooms or entire structures.

PROJECTS

- Spaceship servicing using a **3-DOF robotic arm** (Intelligent Robotics course).
- Comparison of the performance of **Recommendation systems using RBM, GNN, and Transformers**.
- **Obstacle Avoidance and Path Planning Robot** using Deep Reinforcement Learning.
- **Modeling and Simulation of Robotic Arm** using MATLAB.
- Social Distance Management System via video surveillance (**OpenCV**).

TEACHING EXPERIENCE

University of Cincinnati

- **Teaching Assistant – Deep Learning (CS)** Jan 2025 – May 2025
Conducted weekly office hours to support students with key deep learning concepts and implementation challenges. Provided actionable feedback on assignments and projects, improving their technical and conceptual understanding.
- **Teaching Assistant – Automata Theory (CS)** Jan 2025 – May 2025
Held regular and supplemental office hours to help students grasp formal language theory including DFAs, NFAs, and grammars. Supported students in problem-solving and exam preparation through clear explanations and structured examples.
- **Teaching Assistant – Physics Department** Sep 2024 – Dec 2024
Assisted professors as a floating TA during lecture assignments and led recitation sessions simplifying core physics concepts with illustrative examples. Provided cross-section academic support, earning strong end-of-semester feedback and recognition from students for teaching clarity and approachability.

CERTIFICATES

- Machine Learning with Python (cognitiveclass.ai - IBM Developer Skills Network)
- Control of Mobile Robots (Coursera/Georgia Tech)
- Mathematics for Machine Learning: PCA (Coursera/Imperial College of London)
- Structuring Machine Learning Projects (Coursera/DeepLearning.AI)
- Mathematical Thinking in Computer Science (Coursera/UC San Diego)

AWARDS

- Graduate Incentive Award - University of Cincinnati (2024-Current)
- Employee of the Quarter (Jan-Mar '23, Jan-Mar '24).
- Rajeswari Viswanathan Memorial Scholarship (2019 - 2020).
- Won Robowars (under 30kg category) in university (VIT).
- 1st Rank in Junior College (12th Grade) for academic excellence.